

DATA_ENG 300 Homework 3

5.23.26

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1. EC2 Setup

Instance: t2.large, AMI: Amazon Linux 2023, region: us-east-1. Java (OpenJDK Corretto 17.0.10) and PySpark 4.0.2 were installed inside a Python virtual environment. Jupyter was launched on EC2 with a local SSH tunnel on port 8888 for browser access from a Mac.

Data was pulled from S3 using AWS CLI to ~/data/nyctlc/. Results were saved locally then re-uploaded to S3.

```
import subprocess
print(subprocess.run(["java", "-version"], capture_output=True, text=True).stderr)
import pyspark
print(pyspark.__version__)

openjdk version "17.0.10" 2024-01-16 LTS
OpenJDK Runtime Environment Corretto-17.0.10.7.1 (build 17.0.10+7-LTS)
OpenJDK 64-Bit Server VM Corretto-17.0.10.7.1 (build 17.0.10+7-LTS, mixed mode, sharing)

4.0.2
```

2. Dataset

Month used: January 2026. Two Parquet files from the course S3 bucket:

- yellow_tripdata_2026-01.parquet
- green_tripdata_2026-01.parquet

Yellow and green schemas were standardized to 9 common columns (taxi_type, pickup_datetime, dropoff_datetime, pickup_location_id, dropoff_location_id, trip_distance, fare_amount, total_amount, passenger_count) before union.

3. Data Cleaning

The following filters and assumptions were applied:

Filter Rule	Assumption
Remove null pickup or dropoff timestamps	Trips with no time info are unusable
Remove trip_distance <= 0	A trip must cover positive distance
Remove fare_amount < 0	Negative fares are billing errors
Remove total_amount < 0	Same reasoning as fare
Remove dropoff <= pickup time	Data entry error or corruption
Remove trip duration > 24 hours	Likely an unclosed or stale record

4. Analytics Results

Q1: Which taxi type had more trips?

taxi_type	avg_fare
green	16.0
yellow	21.08

Yellow taxis had a lot more trips than green taxis because yellow cabs dominate the city (Manhattan) while green cabs serve the outer boroughs.

Q2: Average fare by taxi type?

taxi_type	count
yellow	3518128
green	38909

Yellow taxis average a higher fare, likely due to more Manhattan and airport routes.

Q3: Average trip distance by taxi type?

taxi_type	avg_distance
green	12.99
yellow	6.76

Green taxis average longer trips. Outer borough trips tend to be longer in miles even if the fare is lower.

Q4: What hour had the most pickups?

pickup_hour	count
18	232570
17	224277
15	216305
19	211118
20	208072
21	207428
16	206485
14	202317
13	190970
22	190251
12	183146
11	167923
10	158111
9	154087
23	148822
8	146020
7	112027
0	109013
1	75623
6	62649
2	52470
3	37491
5	32070
4	27792

Hour 18 (6 PM) had the highest count at 232,570, followed by hours 17 and 15. The busiest window is roughly 12 PM to 10 PM, aligning with afternoon/evening commute and dinner hours. The slowest hours are 4 to 5 AM.

Q5: Percentage of trips under 2 miles?

Trips under 2 miles: 1847954
Total trips: 3557037
Percentage: 51.95%

Just over half of all trips are under 2 miles, short hops are the dominant especially for yellow cabs in dense Manhattan.

5. Fare Prediction Model (Q8)

- **Model:** Random Forest Regressor (80 trees, max depth 8).
- **80/20 train-test split** (seed=42)
- **Predictors:** trip_distance, pickup_location_id, dropoff_location_id, passenger_count.

RMSE Results:

Training RMSE: \$7.39
Test RMSE: \$7.68

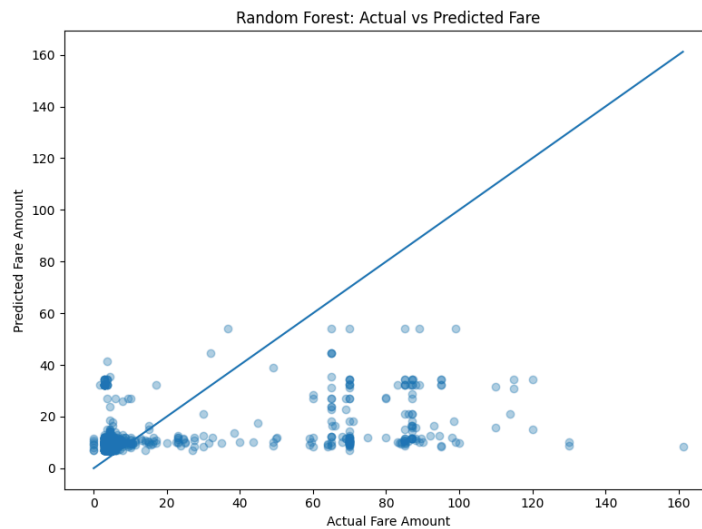
The small gap between training and test RMSE suggests the model generalizes well without significant overfitting.

Feature Importance Analysis:

	feature	importance
0	trip_distance	0.819885
1	pickup_location_id	0.130795
2	dropoff_location_id	0.045709
3	passenger_count	0.003611

- trip_distance was by far the most important predictor, which makes sense since NYC taxi fares are metered primarily by distance.
- pickup_location_id and dropoff_location_id were the next most important features. Certain zones such as airports and midtown consistently produce higher fares due to surcharges and tolls, regardless of distance.
- passenger_count had the lowest importance, which is expected since NYC taxi fares do not scale with number of passengers because one passenger and four passengers pay the same fare for the same trip.

Actual vs Predicted Plot:



The scatter plot shows points forming a tight diagonal band for typical fares (\$0–\$50), indicating good model performance in that range. Spread increases at higher fare values for airport and long-distance trips, where the model slightly underestimates because those trips involve surcharges and tolls that are not captured in the predictor columns.